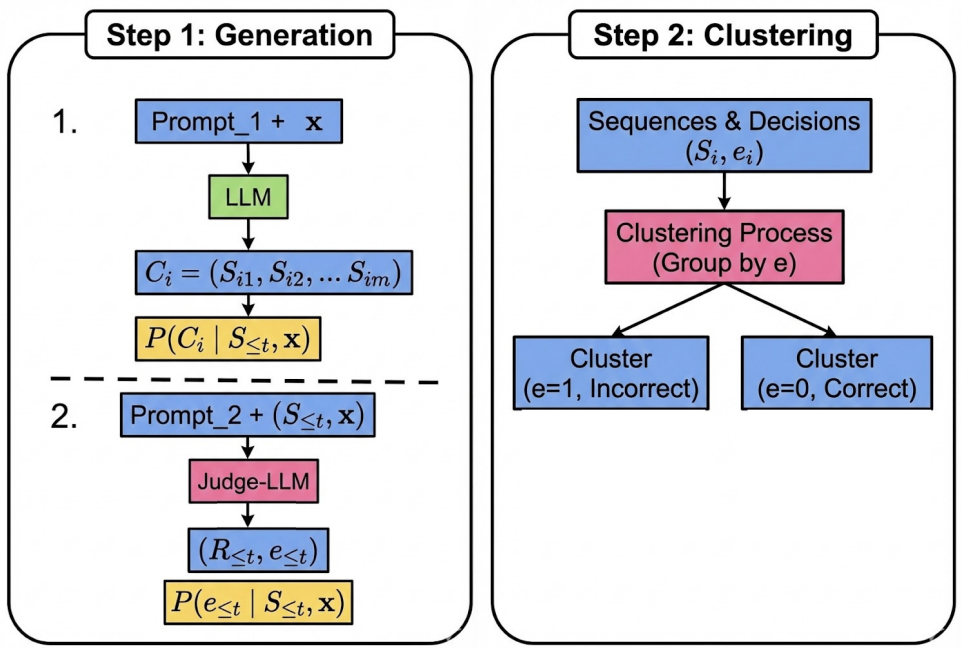
CoT Entropy

CoT Entropy[119]

Main Steps:

1. **Generation:** Given a judge-LM and a context $\mathbf{x}_{\leq t}$, sample reasoning sequences c that critique the current step s_t, followed by a decision e (of whether s_t is deemed incorrect) after the CoT.
2. **Clustering:** Group the output sequences into two clusters based on whether the decision e equals 1 (deemed incorrect) or 0 (deemed correct).
3. **Entropy estimation:** Normalize token probabilities $\tau(e)$ to obtain class probabilities for each sequence in both clusters. Then, sum the class probabilities of sequences leading to the same decision e and compute the resulting entropy.



Step 3: Entropy estimation

$$\mathbf{x}_{\leq t} \coloneqq (Q, \mathbf{s}_{\leq t}, \mathbf{I}_{\text{CoT}}),$$

$$\text{CoTE}(\mathbf{x}_{\leq t}) = -\sum_e \left(\left[\sum_c p_\theta(c, e \mid \mathbf{x}_{\leq t}) \right] \log \left[\sum_c p_\theta(c, e \mid \mathbf{x}_{\leq t}) \right] \right)$$

$$= -\sum_e \left(\left[\sum_c p_\theta(e \mid \mathbf{x}_{\leq t}, c) p_\theta(c \mid \mathbf{x}_{\leq t}) \right] \log \left[\sum_c p_\theta(e \mid \mathbf{x}_{\leq t}, c) p_\theta(c \mid \mathbf{x}_{\leq t}) \right] \right),$$

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CoT Entropy[119]

- **Task Type:** Opening-end/Math/Code generation

- **Introduction:** CoT Entropy, extending semantic entropy, marginalizes chain-of-thought rationales to estimate posterior entropy over step-wise verification decisions. CoT Entropy quantifies reasoning uncertainty, providing a structured probabilistic signal for uncertainty-aware process reward modeling.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Reasoning	White-box	Post-hoc	Low	Ranking

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